STT 442 – Homework 1

Facilitator: Lydia Morey
Reviewer: Ahmad Rubaii
Compiler: Gabriela Villani

Problem 1

(a) The *Facilitator* is designated to initiate and organize discussion among the group to share individual worked answers and general comments on the homework. The *Compiler* is designated to take the worked answers after the discussion and synthesize them into a draft document for the assignment submission. The *Reviewer* is designated to check the draft, noting things like writing and potential inconsistencies from the discussion or inaccuracies and notify the compiler of the issue, or resolve it themselves in the draft for a final submission.

Meaningful contribution is for the group member to fulfill their role, and participate adequately in discussion and general communication.

(b) We will communicate through Zoom, email and the D2L discussion board for the group every few days for check up, and more frequently leading up to the deadline. Members should ideally respond to a message within 24–48 hours.

(c) The group will discuss along the way, from individual work to draft, to final submission, making sure everyone is on board.

(d) A member should notify the group within 24 hours.

(e) The first step will be acknowledgement to the member in a convivial and respectful manner.

Problem 2

(a) We have the expectation and variance as

$$\begin{aligned}
\mathbf{E}[X] &= \int_0^1 xf(x)dx \\
&= \int_0^1 x(3x^2)dx \\
&= \int_0^1 3x^3dx \\
&= 3\left[\frac{x^4}{4}\right]_0^1 \\
&= \frac{3}{4}(1^4) - \frac{3}{4}(0^4) \\
&\boxed{\mu = \frac{3}{4}}
\end{aligned}$$

$$\begin{aligned}
\mathbf{E}[X^2] &= \int_0^1 x^2f(x)dx \\
&= \int_0^1 x^2(3x^2)dx \\
&= \int_0^1 3x^4dx \\
&= 3\left[\frac{x^5}{5}\right]_0^1 \\
&= \frac{3}{5}
\end{aligned}$$

$$\begin{aligned}
\mathrm{Var}(X) &= \mathbf{E}[X^2] - (\mathbf{E}[X])^2 \\
\sigma^2 &= \frac{3}{5} - \left(\frac{3}{4}\right)^2 \\
&= \frac{3}{5} - \frac{9}{16} \\
&= \frac{48}{80} - \frac{45}{80} \\
&\boxed{\sigma^2 = \frac{3}{80}}
\end{aligned}$$

(b) As a basic property of the sample mean, $\mathbf{E}[\bar{X}_{45}] = \mathbf{E}[X] = \frac{3}{4}$. Next,

$$\begin{aligned}
\text{Var}(\bar{X}_{45}) &= \frac{3/80}{45} \\
&= \frac{3}{3600} \\
&= \boxed{\frac{1}{1200}} \\
\text{SE}(\bar{X}_{45}) &= \sqrt{\frac{1}{1200}} \\
&= \boxed{0.0289}
\end{aligned}$$

(c) σ measures the dispersion of individual data points around μ, while the standard error $\frac{\sigma}{\sqrt{n}}$ measures the dispersion of the sample means around μ. As n increases, σ remains constant, but $\frac{\sigma}{\sqrt{n}}$ decreases.

(d) This reasoning is flawed because the observations are independent; future outcomes do not correct past results. If an initial subset has a sample mean below μ, subsequent observations are not forced to be larger. Instead, the Law of Large Numbers simply guarantees that as $n \to \infty$, the overall sample mean converges to μ.

(e) The CLT is applicable because $n = 45 \geq 30$ and the observations are independent, allowing the sample mean distribution to approach normality despite the underlying population.

Calculating the standard error as $\text{SE} = \frac{\sigma}{\sqrt{n}} = \sqrt{\frac{3/80}{45}} \approx 0.02887$, we standardize the sample mean to find the z-score: $z = \frac{\bar{X}-\mu}{\text{SE}} = \frac{0.78-0.74}{0.02887} \approx 1.04$. Therefore:

$$P(\bar{X}_{45} > 0.78) = P(Z > 1.04) = 1 - P(Z \leq 1.04) = 1 - 0.8508 = 0.1492$$

Problem 3

(a) Let T represent the total weight of the 80 packages, such that $T = \sum_{i=1}^{80} X_i$. Given $\mu = 20$ and $\sigma = 15$:

$$\begin{aligned}
\mathbf{E}[T] &= 80(20) = 1600 \\
\text{Var}(T) &= 80(15^2) = 18000 \\
\sigma_T &= \sqrt{18000} \approx 134.164
\end{aligned}$$

Using the normal approximation, the probability that the total weight exceeds 1800 is:

$$P(T > 1800) = P\left(Z > \frac{1800 - 1600}{134.164}\right) = P(Z > 1.49) = 1 - 0.9319 = 0.0681$$

(b) (i) $S \sim \text{Binomial}(90, 0.20)$. The mean is $\mathbf{E}[S] = np = 90(0.20) = 18$, and the variance is $\text{Var}(S) = np(1-p) = 90(0.20)(0.80) = 14.4$.

(ii) Using the CLT without a continuity correction:

$$P(S \leq 10) \approx P\left(Z \leq \frac{10 - 18}{\sqrt{14.4}}\right) = P(Z \leq -2.108) = 0.0175$$

(c) (i) Redoing the approximation with a continuity correction (cutoff 10.5):

$$P(S \leq 10.5) \approx P\left(Z \leq \frac{10.5 - 18}{\sqrt{14.4}}\right) = P(Z \leq -1.976) = 0.0241$$

(ii) We compute the exact probability in R:

```
pbinom(10, size = 90, prob = 0.20)
## [1] 0.01896302
```

The exact binomial probability is 0.01896. Interestingly, the uncorrected approximation (0.0175) is closer to the exact value than the corrected approximation (0.0241), demonstrating that the continuity correction does not uniformly improve precision for all discrete intervals.

Problem 4

(a) (i)

- **Error 1:** "$Var(S) = np = 64(0.25) = 16$"
  The variance of a binomial variable is $np(1-p)$, yielding $\text{Var}(S) = 64(0.25)(0.75) = 12$.
- **Error 2:** *"the Central Limit Theorem tells us that S is exactly normal"*
  The CLT provides only an *approximate* normal distribution for finite n, never an exact one.
- **Error 3:** "$P\left(Z \leq \frac{12-16}{16}\right)$"
  The standard score formula divides by the standard deviation, not the variance. The denominator must be $\sqrt{12}$.
- **Error 4:** *Failing to apply a continuity correction.*
  Approximating a discrete binomial with a continuous normal distribution requires adjusting the bound to $P(S \leq 12.5)$.

(ii) Let $S \sim \text{Binomial}(64, 0.25)$. The mean is $\mu = 16$ and the variance is $\sigma^2 = 12$, yielding a standard deviation of $\sigma = \sqrt{12} \approx 3.464$. Applying the continuity correction:

$$P(S \leq 12) \approx P\left(Z \leq \frac{12.5 - 16}{\sqrt{12}}\right) = P(Z \leq -1.0104) \approx 0.1562$$

The exact probability is verified in R:

```
pbinom(12, size = 64, prob = 0.25)
## [1] 0.1558446
```

The exact probability is 0.1558, demonstrating a highly accurate approximation.

(iii) Dividing by the variance (16) instead of the standard deviation ($\sqrt{12}$) caused the largest numerical error in the final probability. However, claiming the distribution is "exactly normal" is the most serious conceptual error regarding the CLT.

(b) (i) **False.** The variable $(n-1)S^2/\sigma^2$ has a chi-square distribution with $n-1$ degrees of freedom, not n.

(ii) **False.** The sample mean $\bar{X}_n$ is only *approximately* normally distributed for $n \geq 30$. It is strictly normal only if the underlying population distribution is itself exactly normal.

Problem 5

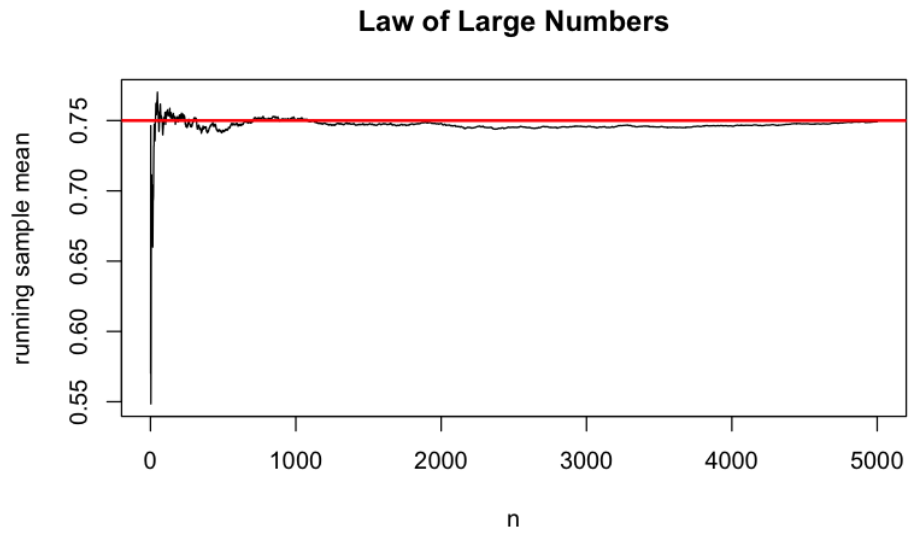


Figure 1: Running sample mean for the $f(x) = 3x^2$ distribution over 5,000 observations.

(a) The running sample mean initially fluctuates significantly before converging toward $\mu = 0.75$. Occasional divergences at larger sample sizes do not contradict the Law of Large Numbers, as the theorem guarantees overall convergence as $n \to \infty$, but allows for random fluctuations along the way. Multiple seeds (e.g., 100, 200, 442) confirm this behavior across distinct random samples.

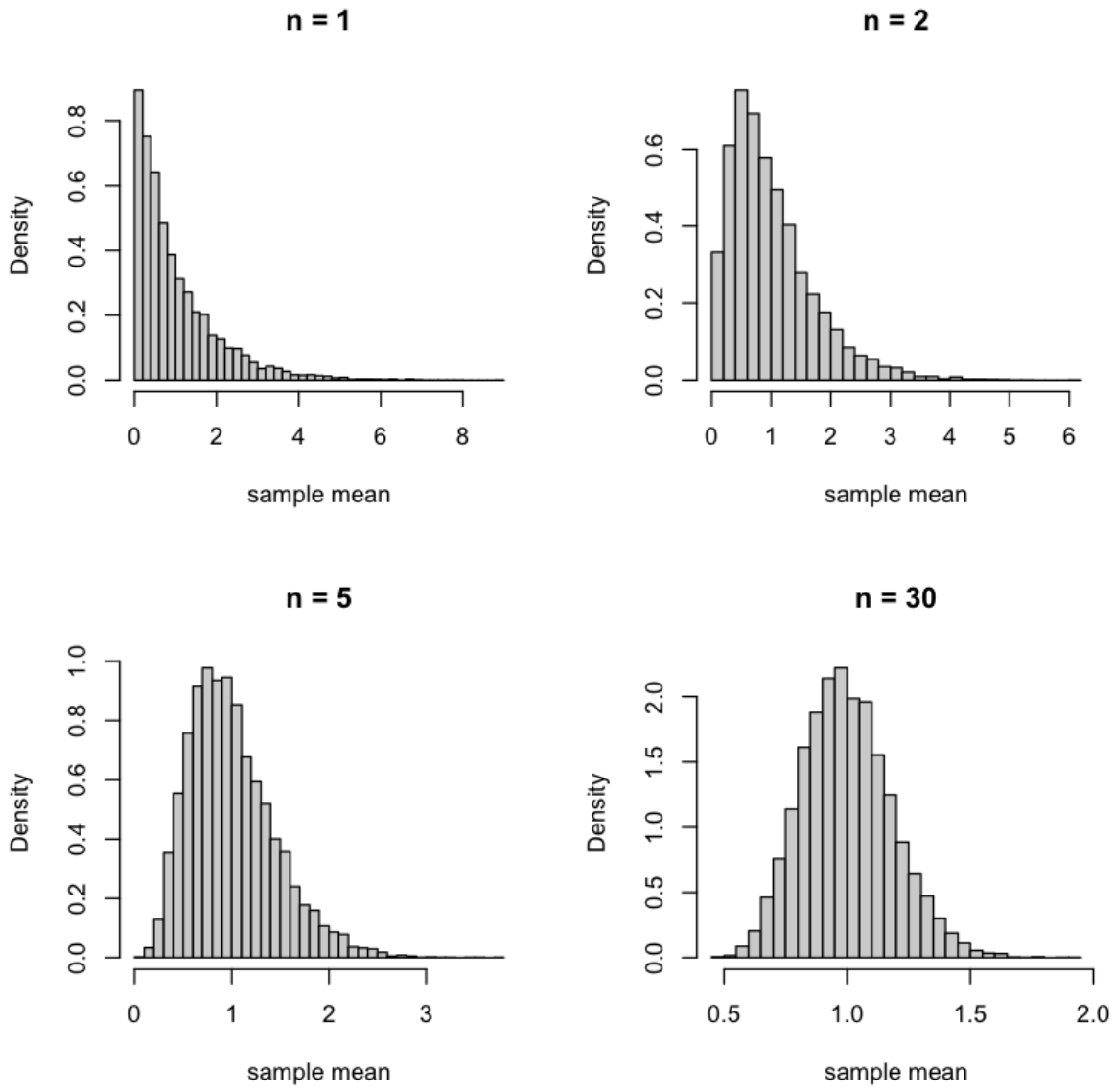


Figure 2: Histograms of sample means from an exponential population for $n \in \{1, 2, 5, 30\}$.

(b) The sampling distribution first appears roughly normal at $n = 30$. For smaller sizes ($n \in \{1, 2, 5\}$), the distributions remain heavily right-skewed, reflecting the underlying exponential distribution.

```
means30 <- sim_means(30)
mean(means30)
## [1] 1.000005
sd(means30)
## [1] 0.1822984
```

At $n = 30$, the simulated mean (≈ 1.000) and standard deviation (≈ 0.182) closely match the theoretical expectations of $\mu = 1$ and $\text{SE} = \frac{1}{\sqrt{30}} \approx 0.1826$.

(c) We simulate from the Bernoulli population ($p = 0.02$) and plot the sample means for $n = 30$ and $n = 500$.

```r
set.seed(442)
sim_bernoulli_means <- function(n, reps = 10000) {
  replicate(reps, mean(rbinom(n, size = 1, prob = 0.02)))
}

par(mfrow = c(1, 2))
hist(sim_bernoulli_means(30), breaks = 40, probability = TRUE,
     main = "n = 30 (Bernoulli p = 0.02)", xlab = "sample mean")
hist(sim_bernoulli_means(500), breaks = 40, probability = TRUE,
     main = "n = 500 (Bernoulli p = 0.02)", xlab = "sample mean")
par(mfrow = c(1, 1))
```

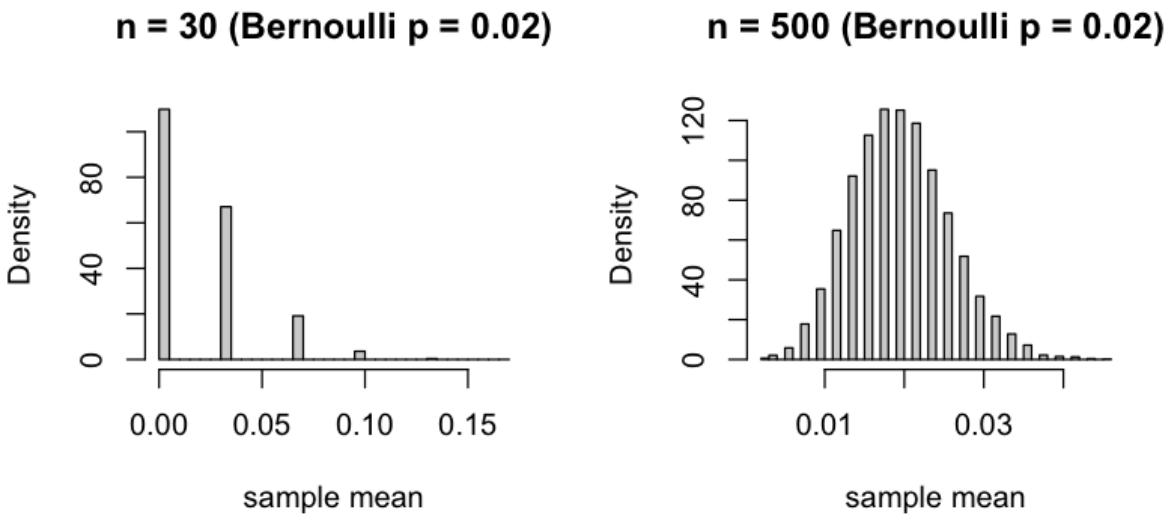


Figure 3: Histograms of sample means from a Bernoulli population for $n = 30$ and $n = 500$.

For this Bernoulli population, the $n = 30$ histogram exhibits large gaps and severe right-skew, departing entirely from normality. A much larger sample size ($n = 500$) is required to achieve a symmetric distribution because successes ($p = 0.02$) are highly infrequent. This demonstrates that the $n \geq 30$ rule of thumb for the CLT is insufficient for highly skewed populations.

(d)

(i) Visualizing the simulations clarified how the sampling distribution of the mean approaches normality as the sample size increases.

(ii) It remains somewhat unclear how to definitively determine the minimum sample size required for a valid normal approximation in arbitrary populations. We compared the histograms across different values of n to better understand this threshold.

Appendix A.

Lydia

(2) a) $\mu = E[X] = \int_{-\infty}^{\infty} x \cdot f(x)\,dx$, $f(x) = 3x^2$, $0 < x < 1$

$\Rightarrow \mu = E[X] = \int_0^1 x(3x^2)\,dx = \int_0^1 3x^3\,dx$

$\Rightarrow \left(\frac{3x^4}{4}\right)\Big|_0^1 = \boxed{\frac{3}{4}} = \mu$

$Var(X) = E(X^2) - E(X)^2$

$E(X^2) = \int_0^1 x^2(3x^2)\,dx = \int_0^1 3x^5\,dx = \frac{3x^5}{5}\Big|_0^1 = \frac{3}{5}$

$E(X)^2 = \left(\frac{3}{4}\right)^2 \Rightarrow Var(X) = \frac{3}{5} - \frac{9}{16} = \boxed{\frac{3}{80}}$

b) $E(\bar{X}_{45}) = E(X) = \mu = \boxed{\frac{3}{4}}$

$Var(\bar{X}_{45}) = \frac{\sigma^2}{n} = \frac{3/80}{45} = \frac{3}{80 \cdot 45} = \boxed{\frac{1}{1200}}$

$SE(\bar{X}_{45}) = \frac{\sigma}{\sqrt{n}} = \sqrt{Var(X)} = \sqrt{1/1200} = \boxed{\frac{1}{\sqrt{1200}}}$

c) σ measures the spread of individual data points in the population and $\sigma/\sqrt{n}$ measures the spread of the sample mean ($\bar{X}_n$). As n increases, σ stays the same and $\sigma/\sqrt{n}$ decreases.

d) Their reasoning is wrong because the observations are independent and future observations don't rely on past observations. The Law of Large Numbers guarantees that as you collect more and more observations they average back to μ which is the true population average

e) You can use CLT here because $45 \geq 30$.

$$Z = \frac{\bar{X} - \mu}{SE} = \frac{0.78 - 0.74}{0.02887} = 1.04$$

$$P(\bar{X}_{45} > 0.78) = P(Z > 1.04) = 1 - P(Z \leq 1.04) =$$

$$= 1 - 0.8508 = \boxed{0.1492}$$

(3) a) $\mu = 20 \quad \sigma = 15 \quad T = 80\,\bar{X}_{80}$

mean $= E(T) = n\mu = 80 \cdot 20 = 1600$ lb
Variance $= Var(T) = n\sigma^2 = 80 \cdot 15^2 = 18000$
$SD(T) = \sqrt{n}\,\sigma = \sqrt{80} \cdot 15 = 134.164$ lb

$P(T > 1800) \qquad \frac{1800 - 1600}{134.164} = 1.49$

$P(T > 1800) = P(z > 1.49) = 1 - P(z \leq 1.49) =$
$= 1 - 0.9319 = \boxed{0.0681}$

b) $S \sim Bin(90, 0.20)$

i) mean $= E(S) = np = 90 \cdot 0.20 = \boxed{18}$
Var$(S) = np(1-p) = 90 \cdot 0.20 \cdot 0.80 = \boxed{14.4}$

ii) $z = \frac{S - \mu}{\sigma} = \frac{10 - 18}{14.4} = -2.11$

$P(S > 10) = P(z > -2.11) = \boxed{0.017}$

c) $z = \frac{10.5 - 18}{\sqrt{14.4}} = -1.98$

i) $P(S \leq 10\,S) = P(z \leq -1.98) = 0.0239$

ii) without continuity correction: 0.0175
with correction: 0.0241 Exact: 0.0216
The correction improves the approximation by a lot.

(4) a)

i) Var(S) = np = 64(0.25) = 16 is wrong

Var(S) = np(1-p) = 64(.25)(.75) = 14

n·p is expected value, not variance

The CLT tells us that S is exactly normal *wrong

The CLT tells us that S is approximately normal. *right

The CLT doesn't give an exact normal distribution, only an approximate

It says the variance is 16, but it is actually 12

$P(z \leq \frac{12-16}{16})$ * wrong

$P(z \leq \frac{12-16}{\sqrt{12}})$ * right

The denominator is supposed to be the standard deviation

Last line says 40.13%, but the right answer is 15.62%

ii) $S \sim Bin(64, 0.25)$

mean: $\mu = np = 64 \cdot 0.25 = 16$

$Var(S) = \sigma^2 = np(1-p) = 64 \cdot 0.25 \cdot 0.75 = 12$

$SD = \sigma = \sqrt{12}$

Since $np = 16 \geq 10$ and $n(1-p) = 48 \geq 10$, we can approximate S with a Normal distribution $N(16, 12)$

$P(S \leq 12) \approx P(S \leq 12.5) = P\left(z \leq \frac{12.5 - 16}{\sqrt{12}}\right) =$

$= P(z \leq -1.0104) =$ **0.1562 or in R: 0.1509**

iii) I think the biggest mistake was dividing by the variance instead of the standard deviation which changed the probability by a lot.

b) i) This statement is wrong. It should say n-1 degrees of freedom. This is because we don't know the value of μ

ii) The use of the word exactly is wrong, it should say approximately

(5) a) The mean gets closer to 0.75 as n increases. Over short intervals the running mean gets farther from μ. This does not contradict the Law of Large numbers because the plot is doing exactly what it's supposed to. (Get closer to μ).

b) The histogram at $n=30$ looks the most normal. At $n=1, 2, 5$, the histogram is skewed very far to the right

mean: $\mu = \frac{1}{\lambda} = \frac{1}{1} = 1$

SD: $\frac{\sigma}{\sqrt{n}} = \frac{1}{\sqrt{30}} = 0.1823$

These match what the theory predicts

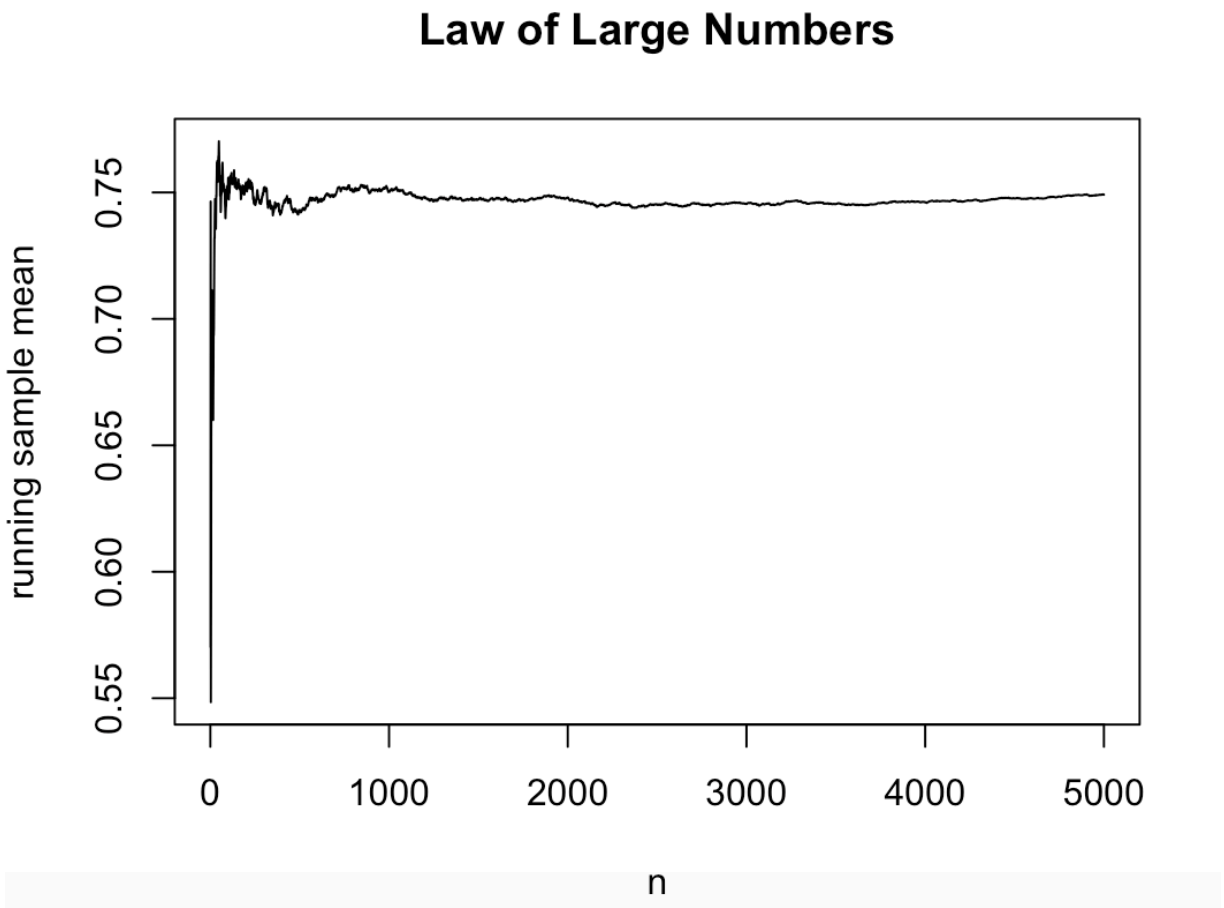
c) At $n=30$, there are large gaps and it is skewed to the right. You can't get a general idea. At $n=500$, the distribution is smooth and easy to read. We need a larger sample size because successes are rare and 30 is too small of a number to see successes. This makes sense with the CLT because as n is going to ∞, the sample mean is looking more normal.

```r

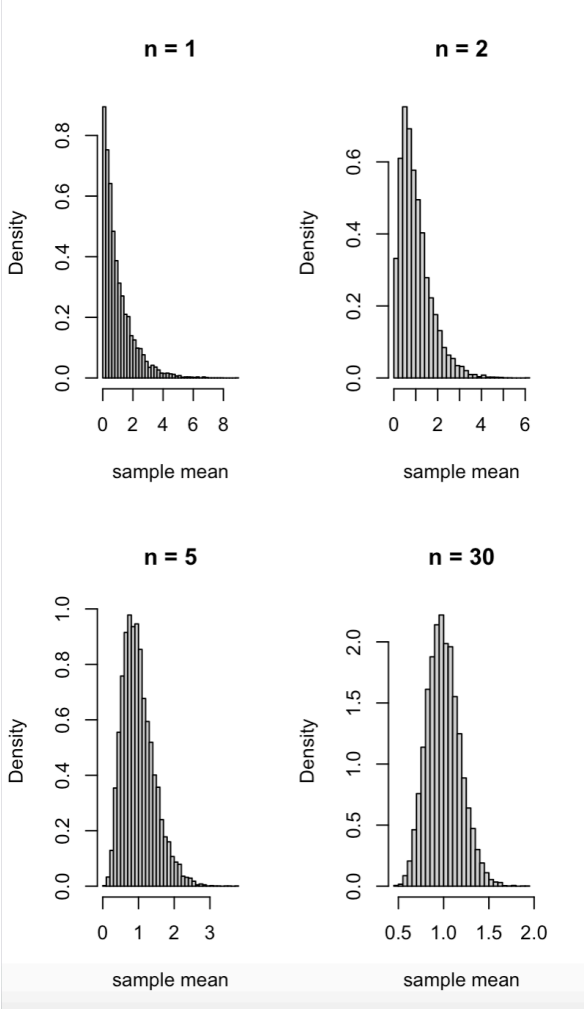
set.seed(442)
x <- runif(5000)^(1/3)
running_mean <- cumsum(x) / seq_along(x)
mu <- 0.75  # Population mean from Problem 2(a)

plot(running_mean, type = "l",
     xlab = "n", ylab = "running sample mean",
     main = "Law of Large Numbers")
abline(h = mu, col = "red", lwd = 2)

```



```r
set.seed(442)
sim_means <- function(n, reps = 10000) {
  replicate(reps, mean(rexp(n, rate = 1)))
}
par(mfrow = c(2, 2))
for (n in c(1, 2, 5, 30)) {
  hist(sim_means(n), breaks = 40, probability = TRUE,
       main = paste("n =", n), xlab = "sample mean")
}
par(mfrow = c(1, 1))
```



```r
set.seed(442)

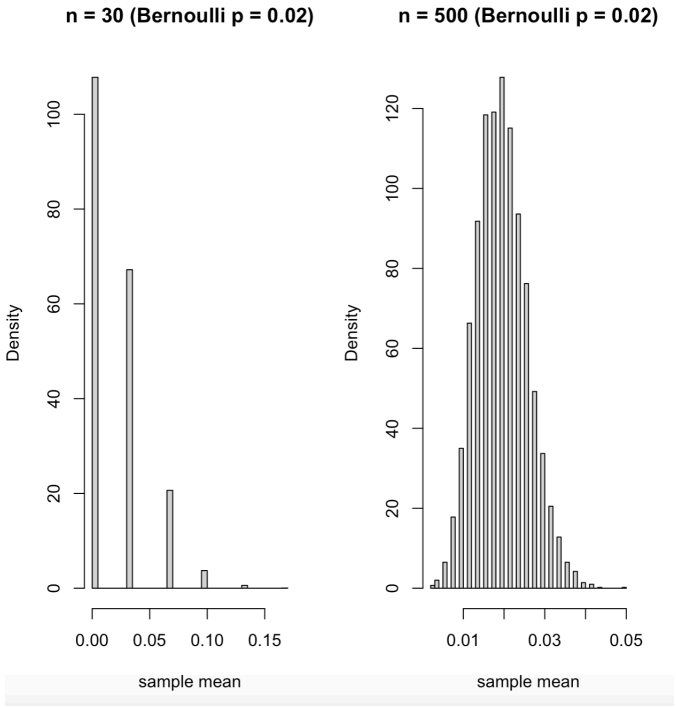
sim_bernoulli_means <- function(n, reps = 10000) {
  replicate(reps, mean(rbinom(n, size = 1, prob = 0.02)))
}

par(mfrow = c(1, 2))


hist(sim_bernoulli_means(30),
     breaks = 40,
     probability = TRUE,
     main = "n = 30 (Bernoulli p = 0.02)",
     xlab = "sample mean")

hist(sim_bernoulli_means(500),
     breaks = 40,
     probability = TRUE,
     main = "n = 500 (Bernoulli p = 0.02)",
     xlab = "sample mean")

par(mfrow = c(1, 1))
```



d) i) I think after seeing the histograms, the theorems are making more sense. The difference between $n=30$ and $n=500$ is huge.

ii) I am still a little confused on how to decide what sample size to use.

Gabriella

2a:

2)

a. $E[X] = \int_0^1 xf(x)dx$

$= \int_0^1 x(3x^2)dx$

$= \int_0^1 3x^3 dx$

$= 3\left[x^4/4\right]_0^1$

$= 3/4(1^4) - 3/4(0^4)$

$\boxed{\mu = 3/4}$

$E[X^2] = \int_0^1 x^2 f(x)dx$

$= \int_0^1 x^2(3x^2)dx$

$= \int_0^1 3x^4 dx$

$= 3\left[x^5/5\right]_0^1$

$= 3/5$

$Var(X) = E[X^2] - (E[X])^2$

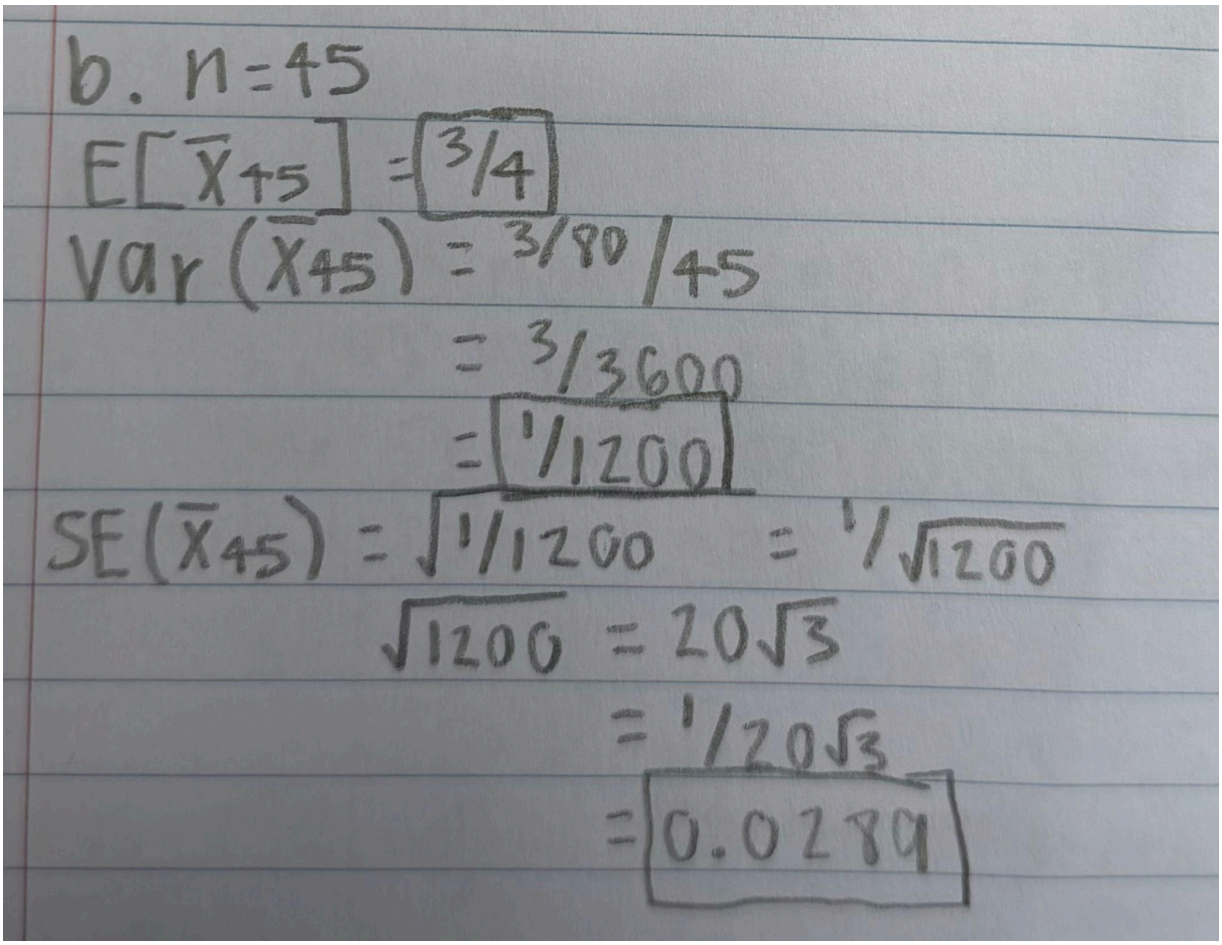
$\sigma^2 = 3/5 - (3/4)^2$

$= 3/5 - 9/16$

$= 48/80 - 45/80$

$\boxed{\sigma^2 = 3/80}$

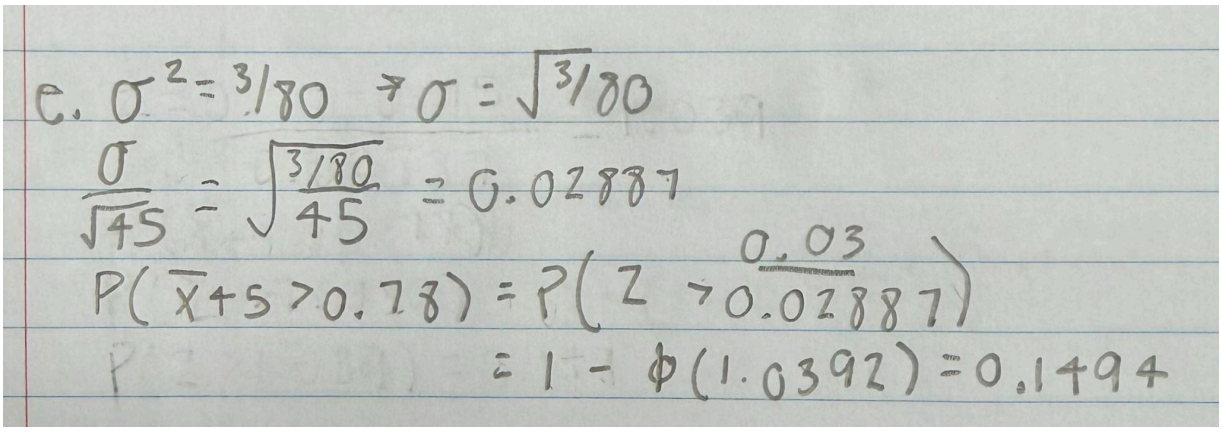
2b:



2c: Sigma represents the standard deviation of the population; hence, it indicates the dispersion of individual data points around the population mean. (sigma/n) represents the standard error, which indicates the dispersion of the sample means around the population mean. The value of sigma does not change as n increases, but the standard error decreases.

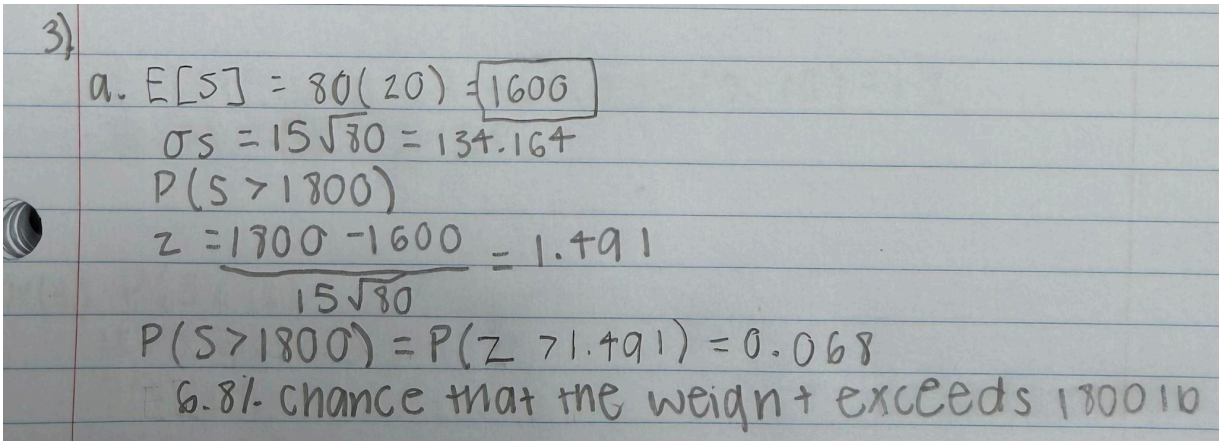
2d: This is not the case because the following observations have no knowledge of what happened before them, and thus do not attempt to correct previous results. If the first 45 observations have a mean less than the true mean, it doesn't necessarily follow that the following observations would have a larger mean. All that the Law of Large Numbers tells us is that the more the number of observations, the closer the sample mean gets to the population mean.

2e:

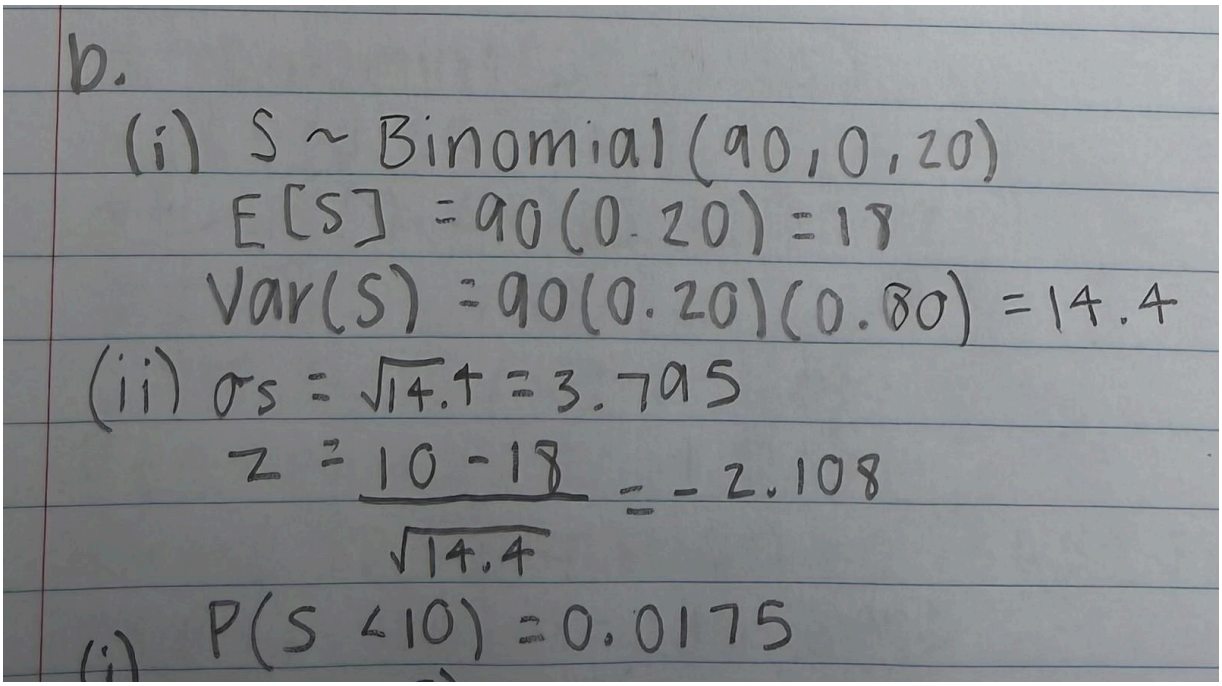


The Central Limit Theorem applies in this case due to the fairly large sample and independence of observations. Even though the underlying distribution of the data is not normal, the distribution of the sample mean approaches normality as the sample size grows.

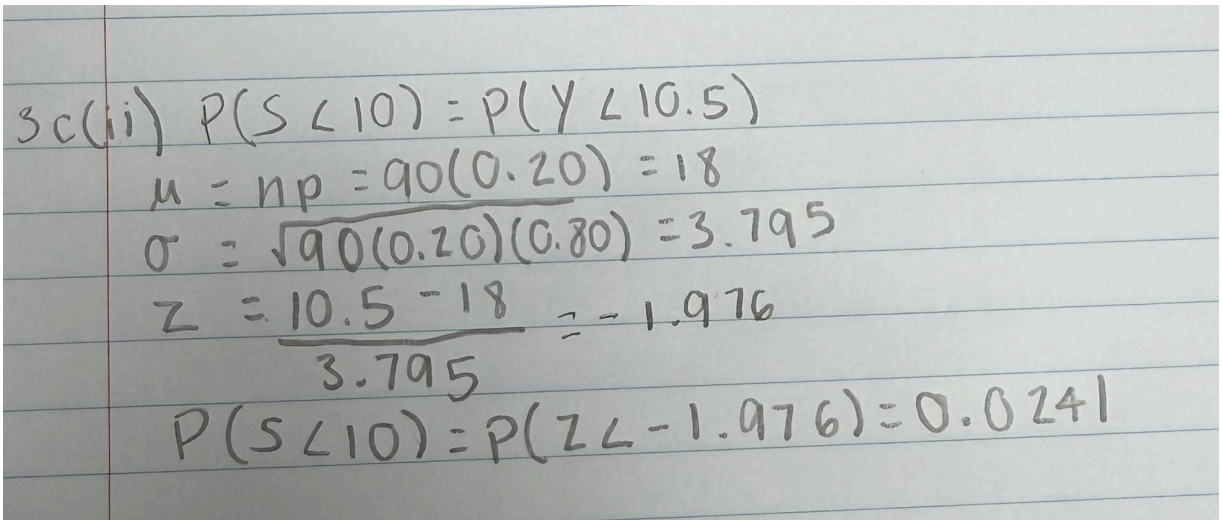
3a:



3b:


b.
(i) S ~ Binomial(90, 0.20)
E[S] = 90(0.20) = 18
Var(S) = 90(0.20)(0.80) = 14.4
(ii) σS = √14.4 = 3.795
z = (10 − 18) / √14.4 = −2.108
(i) P(S < 10) = 0.0175

3c(i):


3c(ii) P(S < 10) = P(Y < 10.5)
μ = np = 90(0.20) = 18
σ = √90(0.20)(0.80) = 3.795
z = (10.5 − 18) / 3.795 = −1.976
P(S < 10) = P(Z < −1.976) = 0.0241

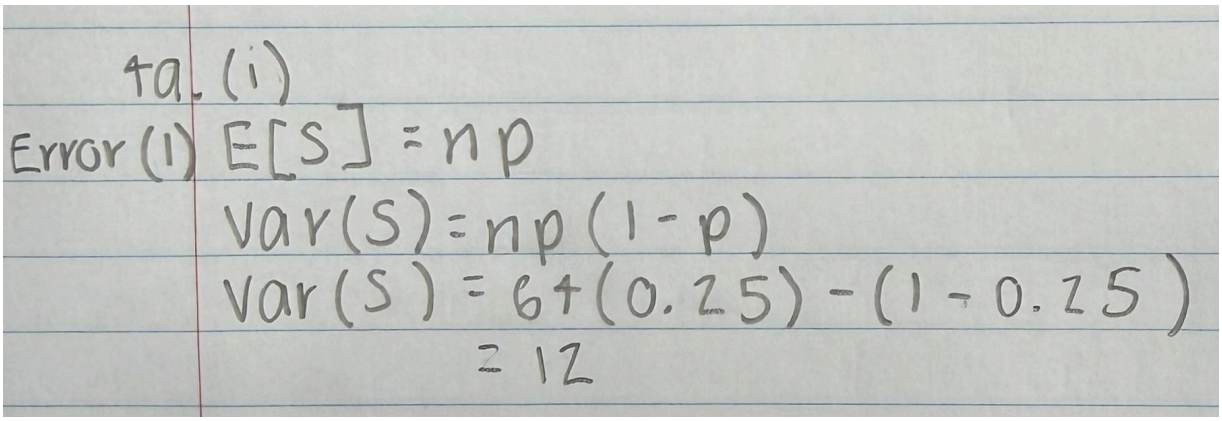
3c (ii):

```
3 c(ii)
```{r}
pbinom(10, size = 90, prob = 0.20)
```
```

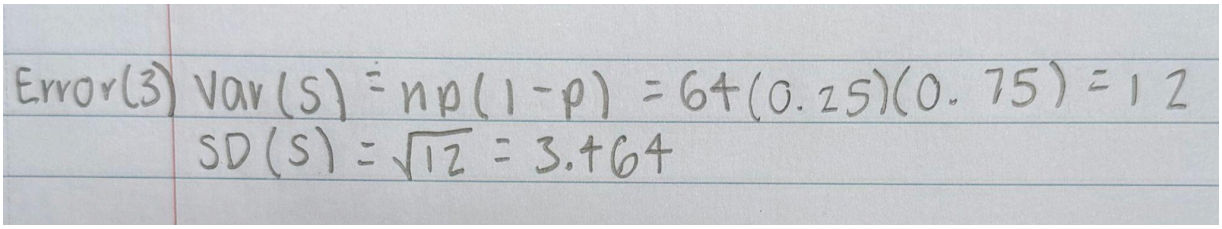
```
[1] 0.01896302
```

The approximation without using the continuity correction turns out to be closer to the actual solution in this example. The continuity correction is not always a guarantee of precision.

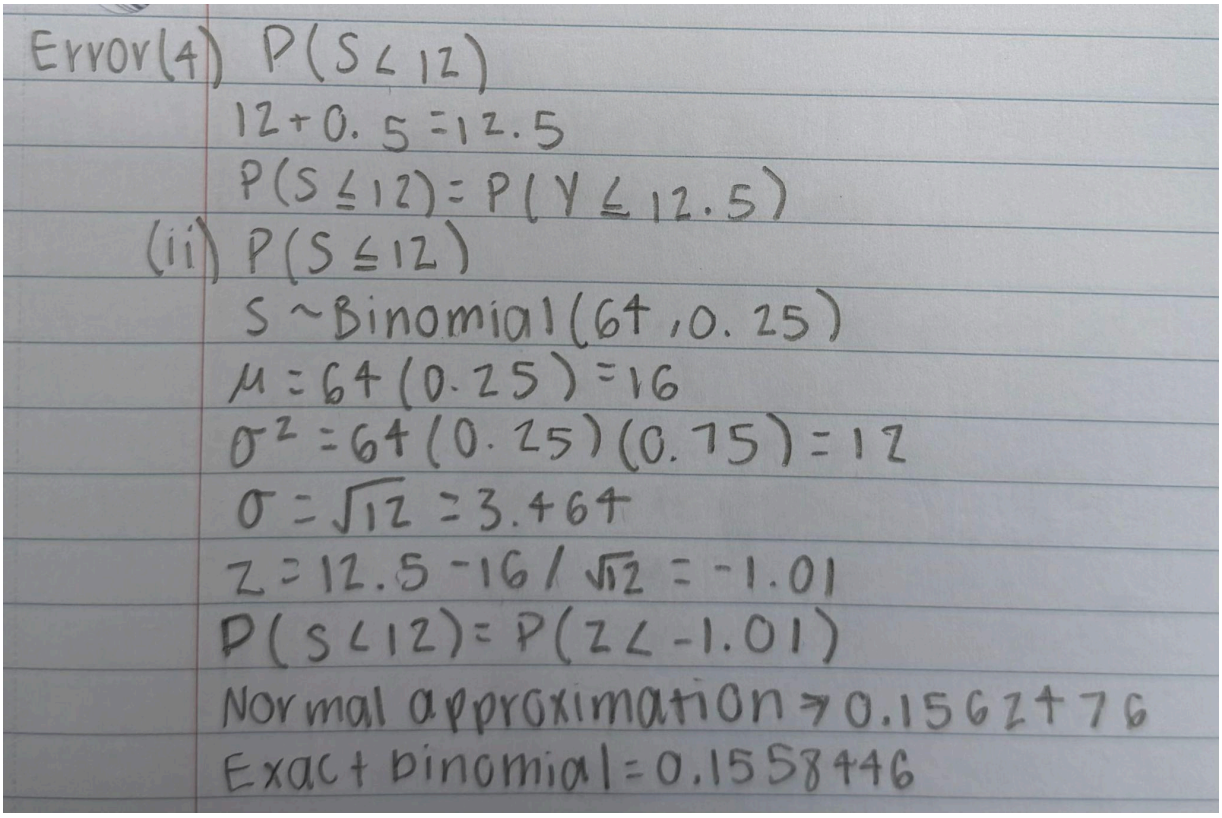
4a: (i)



Error (1): In the problem solved by the student, they assumes that the variance is np=16. This is not correct since the variance equals np(1-p).
Error (2): The student says that S is exactly normal because n>30. This is wrong because the CLT only gives us an approximation to a normal distribution, not an exact normal distribution.
Error(3):



The student applied 16 in the denominator in z formula. This is incorrect because the denominator must be the standard deviation, and not the mean/variance.
Error(4):



The student did not make any continuity correction. As we are approximating a discrete probability distribution by a continuous one, we need to replace 12 by 12.5.
error(5): The student claims that the probability is 40.13%. However, this is not true since the normal distribution is only used as an approximation of the binomial distribution and the exact probability should be calculated using the binomial distribution.

4(a)(ii)

```{r}
pnorm(-1.01)
```

```
[1] 0.1562476
```

4(a)ii

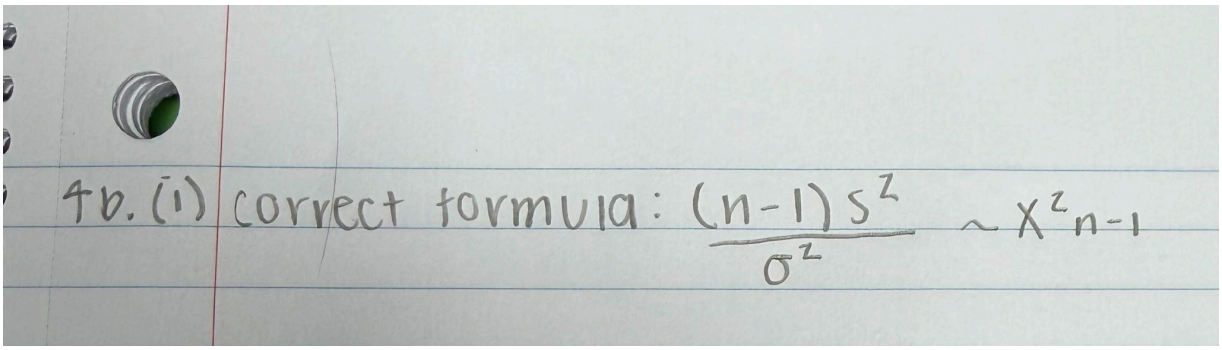
```{r}
pbinom(12, size = 64, prob = 0.25)
```

4a (ii): [1] 0.1558446

4. A (iii) However, the greatest error which affects the numeric result is the use of 16 instead of square root of 12 in the denominator of the z-value. This is because it affected the probability to a great extent. In my opinion, the greatest error in this case is the one with the Central Limit Theorem.

4b.

(i)

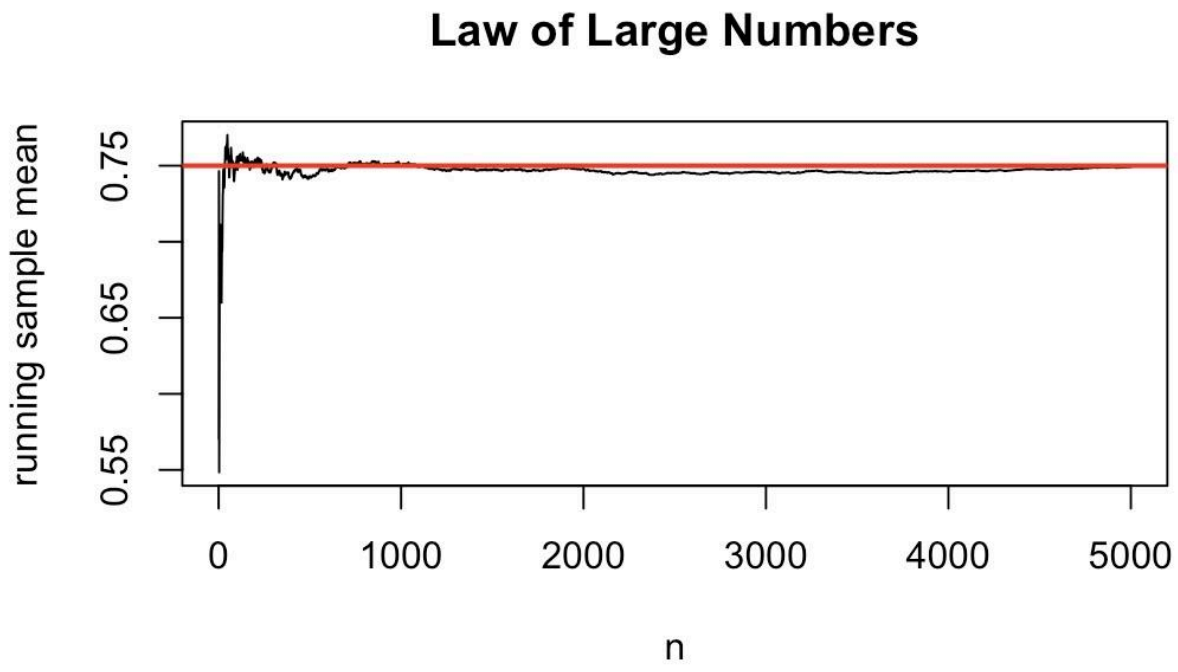


(ii) His statement is wrong because according to the central limit theorem, as the size of the samples increases, their means tend to become normally distributed but it does not imply that they will always be normally distributed for n>30. Thirty is just an empirical figure, which may not apply to all distributions.

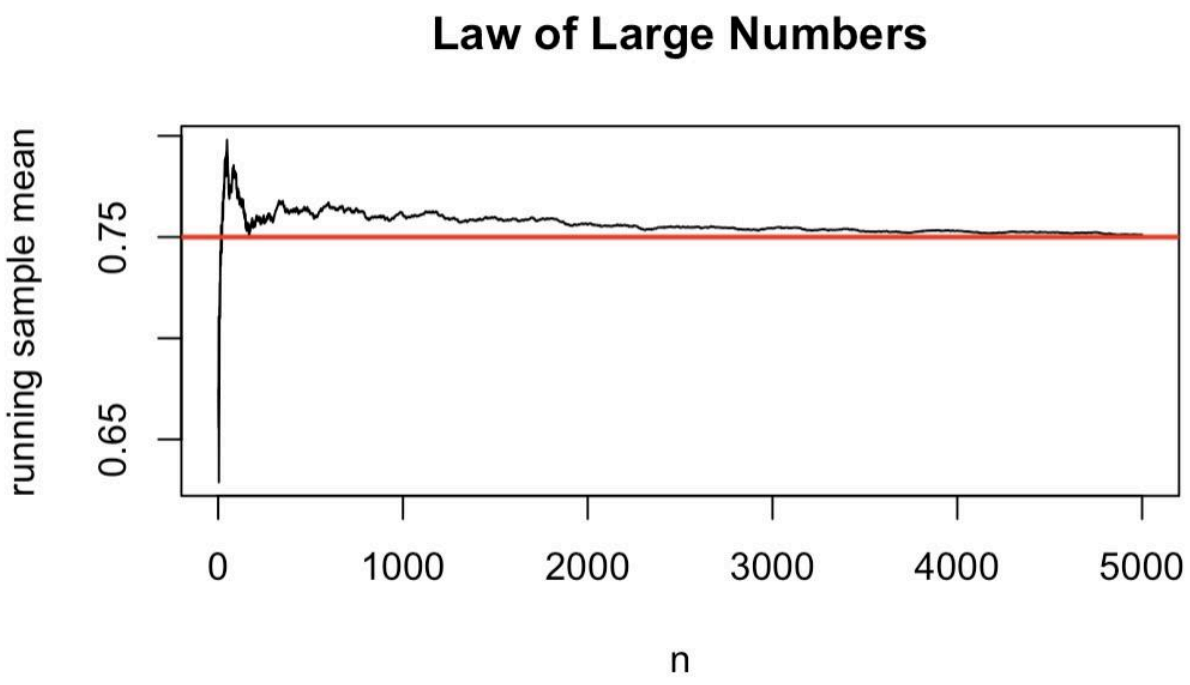
5(a):

5(a)

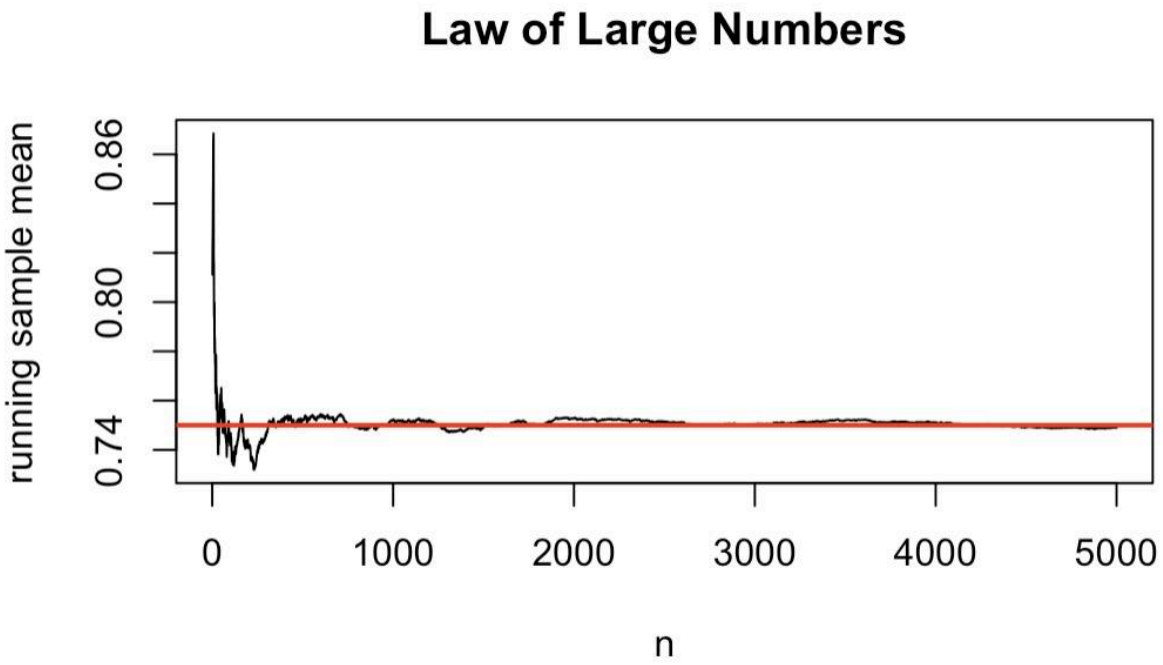
```{r}
set.seed(442)
x <- runif(5000)^(1/3)
running_mean <- cumsum(x) / seq_along(x)
mu <- 0.75
plot(running_mean, type = "l",
     xlab = "n", ylab = "running sample mean",
     main = "Law of Large Numbers")
abline(h = mu, col = "red", lwd = 2)
```



```{r}
set.seed(100)
x <- runif(5000)^(1/3)
running_mean <- cumsum(x) / seq_along(x)
mu <- 0.75
plot(running_mean, type = "l",
     xlab = "n", ylab = "running sample mean",
     main = "Law of Large Numbers")
abline(h = mu, col = "red", lwd = 2)
```



```r
set.seed(200)
x <- runif(5000)^(1/3)
running_mean <- cumsum(x) / seq_along(x)
mu <- 0.75
plot(running_mean, type = "l",
     xlab = "n", ylab = "running sample mean",
     main = "Law of Large Numbers")
abline(h = mu, col = "red", lwd = 2)
```

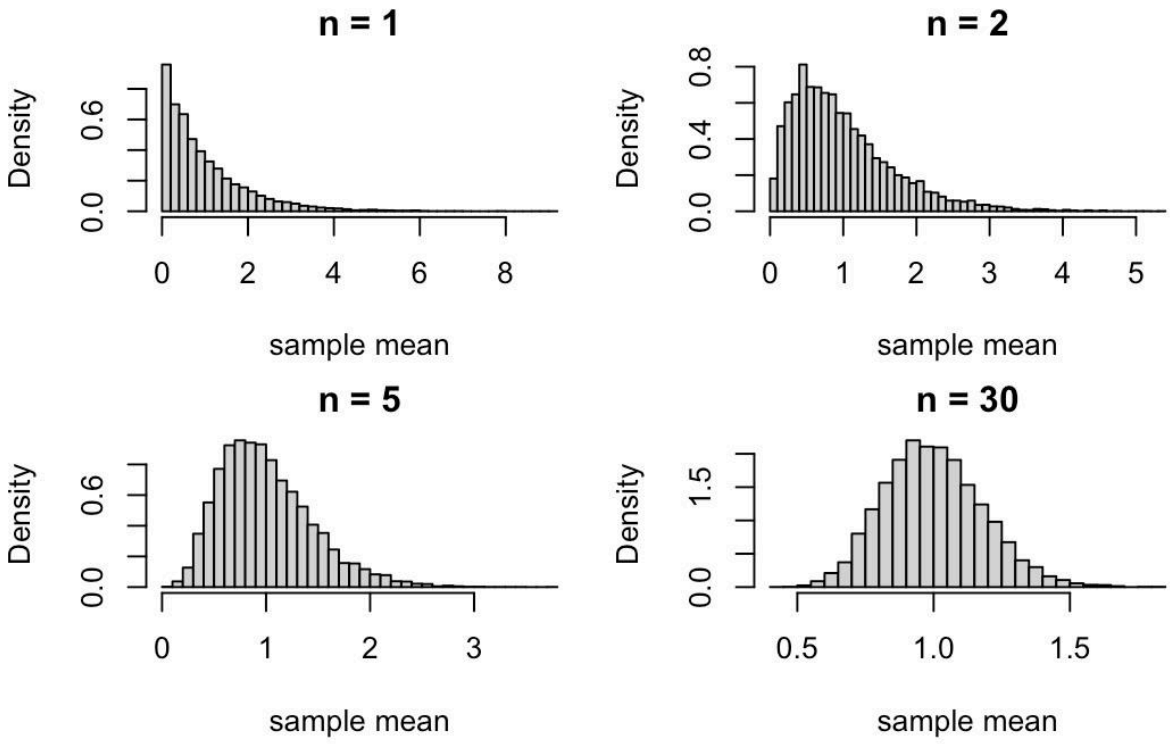


However, at the initial stage, the running sample mean fluctuates a great deal; however, as the sample size becomes larger, it begins to converge on the actual value of the mean (0.75). At some times, however, it may diverge from the mean as n increases. This is not an infringement of the Law of Large Numbers since this law states that the sample mean converges to the actual mean value as n becomes large; it does not have to converge all the time.

5(b):

5(b)

```{r}
par(mfrow = c(2, 2))
par(mar = c(4, 4, 2, 1))
for (n in c(1, 2, 5, 30)) {
  hist(sim_means(n), breaks = 40, probability = TRUE,
       main = paste("n =", n), xlab = "sample mean")
}
par(mfrow = c(1, 1))
```



```{r}
set.seed(442)
means30 <- sim_means(30)
mean(means30)
sd(means30)
```

```
[1] 1.000005
[1] 0.1822984
```

The histogram appears to be normally distributed first for n = 30. In case of n = 1 and n = 2, the histograms appear highly skewed to the right, while for n = 5, the distribution is somewhat closer to being normally distributed. For n = 30, the mean from simulation is 1.000005 and the standard deviation is 0.1822984. These are quite similar to theoretical mean 1 and standard deviation 1 over the square root of 30 equals 0.1826.

5(c): At n=30, the histogram is extremely right-skewed, and it is not normally distributed due to the very low value of p = 0.02. At n=500, the histogram appears to be much more symmetric. In general, this population requires a larger sample size since the probability of the event is quite low. According to the CLT, the sample mean tends to become normally distributed when n approaches infinity.

5(d)

(i) What is one thing you understand better after doing the simulation?

The Central Limit Theorem is clearer now that I have seen the simulations done. I know how the sampling distribution of the mean becomes normally distributed when the size of the samples gets bigger.

(ii) What is still unclear or unresolved? What did you try to resolve it?

I am a bit confused with the minimum sample size that will make the normal approximation good. I compared all the graphs in order to understand their differences.

Ahmad

2a) $$E[X] = \int_0^1 x(3x^2)\,dx$$

$$= \int_0^1 3x^3\,dx$$

$$= \frac{3}{4}\left[x^4\right|_0^1$$

$$= \boxed{\frac{3}{4}}$$

$$E[X^2] = \int_0^1 x^2(3x^2)$$

$$= \frac{3}{5}\left[x^3\right|_0^1$$

$$= \boxed{\frac{3}{5}}$$

$$Var(X) = E[X^2] - (E[X])^2$$

$$= \frac{3}{5} - \frac{9}{16}$$

$$= 3/80 = \sigma^2$$

b) $E[X] = E[\bar{X}_{45}]$

$$Var(\bar{X}_{45}) = \frac{3/80}{45} = \frac{3}{3600} = \frac{1}{1200}$$

$$SE(\bar{X}_{45}) = \sqrt{1/1200} = 0.289$$

2c).

- σ shows the spread of points around the population mean.
- $\sigma/\sqrt{n}$ represents the quantified error between the sample means and the population mean, μ.

2d) This statement is partially true in that the LLN will eventually converge to μ, however the independence of the observations does not guarantee the behavior of the next observations after 45.

2e) $\sigma^2 = 3180 \Rightarrow \frac{\sigma}{\sqrt{45}} = \sqrt{\frac{3180}{45}} = 0.02887$

$$\Rightarrow P(\bar{X}_{45} > 0.78) = P\left(Z > \frac{0.03}{0.02887}\right)$$

$$= 1 - \Phi(1.0392)$$

$$= 0.1494$$

3a) $E[S] = 1600$

$\sigma_S = 15\sqrt{80}$

$Z = \frac{1800 - 1600}{15\sqrt{80}} = 1.491$

$P(S > 1800) = P(Z > 1.491) = 0.068$

6.8% chance.

3b) (i) $S \sim Binomial(90, 0, 20)$

$E[S] = 90(20) = 18$

$Var(S) = 90(.2)(.8) = 14.4$

(ii) $\sigma_S = \sqrt{Var(S)} = 3.795$

$Z = -2.108$

$P(S < 10) = 0.0241$

4a) The first error was

$Var(S) \neq np$

Second error was that $S > 0$ is exactly normal.

Appendix B: Process Notes

Facilitator's Collaboration Summary (Lydia)

Our group worked together by first having each member try all five problems and share their solo work. We then shared our answers to compare our solutions, walk through each problem step by step. Throughout the collaboration process, everyone had the opportunity to present their reasoning. We didn't really disagree on anything specifically, but comparing our answers helped us understand the parts we didn't get at first. Especially problem 4 we had to combine our answers together. By talking through each step as a group, we reached a final consensus on what the answers should be.

Compiler's Compilation Note (Gabriella)

After the group reached a consensus on the solutions, the reconciled work was assembled into a single organized document using LaTeX. Proper formatting was ensured for all mathematical equations, R code chunks, and graphs to make them easy to read.

Reviewer's Change Note (Ahmad)

Once the document was compiled, the entire draft was checked for completeness and correctness. A few minor writing typos and mathematical errors were caught and revised to match the agreed upon solutions. The R code was also double checked to confirm it ran correctly. After verifying that everything looked good, the final changes were made and the file was submitted to D2L.